MATH 231: Linear Algebra I

Section 10 · Fall 2026

Department of Mathematics · Queens College, City University of New York

Instructor	Kevin Bedoya
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Meetings	Tuesdays & Thursdays, 12:10–2:00 PM
Location	Kiely Hall 326
Mode	In-person
Office hours	*to be announced at the first class meeting and posted on the class Discord*
Office location	Kiely Hall 331 (the Math Lab)
Credits	4 hours; 4 credits
Prerequisite	MATH 141 or MATH 151, minimum grade C−
Semester	August 28 – December 21, 2026

Email policy

Write from your QC email account and put `MATH 231` in the subject line. I answer within one business day on weekdays; replies over the weekend are likely but not promised. Questions that other students would benefit from — about homework, a definition, an assignment deadline — are better asked in the class Discord, where everyone can see the answer.

1. Course description

Catalog description. *An introduction to linear algebra with emphasis on techniques and applications. Topics to be covered include solutions of systems of linear equations, vector spaces, bases and dimension, linear transformations, matrix algebra, determinants, eigenvalues, and inner products. Not open to students who are enrolled in or who have completed MATH 237. Students who fail or withdraw from this course multiple times may be prohibited from majoring in the sciences or mathematics; see the bulletin language for your major.*

How this section is taught. Most students in this room are computer science majors or in applied-mathematics-adjacent programs, and the section is built accordingly. We cover the full catalog content, and we lean into the computational side of it: how much a calculation actually costs, how a machine really stores a number, and how the standard algorithms of data science are built out of the vector and matrix operations we develop. Along the way we will meet k-means clustering, k-nearest neighbours, the Gram–Schmidt process, gradient descent, Google's PageRank, and image compression.

On programming. Our language is Python, used for homework, projects and extra credit. **Your programming ability will not be assessed on exams.** If you have never written a line of code, assignments come with walkthroughs covering the syntax and machinery you need; if you have taken CSCI 111, 211 or 212, you will be comfortable quickly. Projects are implementation-based: you will be asked to implement an algorithm that has already been developed in lecture. We will use `numpy`, `scipy` and `matplotlib`.

2. Pathways learning outcomes

MATH 231 satisfies the CUNY Pathways **Mathematical and Quantitative Reasoning** (MQR) requirement, and must meet all six MQR outcomes. Each is stated below in its official wording, followed by what it looks like concretely in this course and where it is assessed.

MQR 1. *Interpret and draw appropriate inferences from quantitative representations, such as formulas, graphs, or tables.*

> Read a matrix as a table of coefficients and recover the linear system it encodes; read an operation-count table and a growth plot to decide which of two algorithms scales better; read a table of distances to classify a point; read the spacing diagram of a floating-point system to say what precision is available near a given magnitude.
>
> *Assessed by: homework, exams.*

MQR 2. *Use algebraic, numerical, graphical, or statistical methods to draw accurate conclusions and solve mathematical problems.*

> Solve linear systems by elimination; compute dot products, norms, projections, determinants and eigenvalues; run the Gram–Schmidt process to produce an orthonormal basis; execute Lloyd's algorithm and the k-nearest-neighbours rule both by hand and in code.
>
> *Assessed by: homework, projects, exams.*

MQR 3. *Represent quantitative problems expressed in natural language in a suitable mathematical format.*

> Turn "find the coordinates of this vector against this basis" into the matrix equation $V\mathbf{x} = \mathbf{u}$; turn a described collection of objects into feature vectors in $\mathbb{R}^n$; turn a stated set of simultaneous constraints into a linear system and its matrix.
>
> *Assessed by: homework, projects, exams.*

MQR 4. *Effectively communicate quantitative analysis or solutions to mathematical problems in written or oral form.*

> Written homework solutions that justify each step rather than only stating an answer; a written report accompanying each project explaining the method, the results, and the limits of both; explaining your reasoning aloud in class and in office hours.
>
> *Assessed by: homework, projects.*

MQR 5. *Evaluate solutions to problems for reasonableness using a variety of means, including informed estimation.*

Substitute a computed solution back into the original system to check it; use a Big-O estimate to judge in advance whether a computation is feasible at a given input size; use relative error and machine precision to decide how many digits of a numerical answer can be trusted; sanity-check a clustering against what the data plainly looks like.

Assessed by: homework, projects, exams.

MQR 6. *Apply mathematical methods to problems in other fields of study.*

Apply linear algebra to problems in computer science and data science: nearest-neighbour search and recommendation, clustering, image compression, PageRank, and the least-squares fitting that underlies regression.

Assessed by: projects.

3. Materials

No purchases are required. Free online copies of every text are linked below and collected in the course GitHub repository. Lectures do not follow any single textbook — I have written my own progression for this course. The texts supply additional reading, alternate perspectives, and problems.

Text	**Role in the course**	**Free access**
Boyd & Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (Cambridge, 2018)	Applied spine	`https://web.stanford.edu/~boyd/vmls/vmls.pdf`
Trefethen & Bau, *Numerical Linear Algebra* (SIAM, 1997)	Numerical methods, conditioning, QR, SVD	`https://www.stat.uchicago.edu/~lekheng/courses/309/books/Trefethen-Bau.pdf`
Strang, *Linear Algebra and Its Applications*, 4th ed. (Cengage, 2006)	Subspaces and geometric intuition	Linked in the course repository
Axler, *Linear Algebra Done Right*, 4th ed. (Springer, 2024)	Theoretical foundation	`https://linear.axler.net`

3.1. Foundations primer — read this first

A self-study primer, *MATH 231 Foundations Review*, is posted in the course repository under `lectures/primer/`. It contains no new linear algebra: it collects the algebra, notation, summation and calculus you already have, so that we can speak a common language on day one. **Read it before or during the first week.** If a page feels like review, skim it; if a page feels new, slow down and work the examples with a pencil.

3.2. Platforms and setup

Everything below is free.

Discord	Primary communication. **All announcements go through the class Discord** — schedule changes, homework, extra credit, projects, exams, and general information. Invite link distributed by email before the first meeting.
GitHub	Lecture notes, homework, projects, practice exams, textbook links: https://github.com/KevinB88/MATH231-Queens-College-Fall-2026-Linear-Algebra-I-
Gradesly	Where grades are posted: https://gradesly.com/. You will sign up with your CUNY ID and a randomly generated passcode; setup instructions are distributed in week 1.
Git + GitHub account	How you submit work. You will push homework, projects and extra credit to your own MATH 231 repository. A setup walkthrough — creating an account, installing git, structuring your directory, and what git actually is — is provided in week 1.
Python + an IDE	VS Code or PyCharm. Both are free.

We are not using Brightspace

Course materials, announcements and grades live on GitHub, Discord and Gradesly respectively. Do not wait for a Brightspace post; you will miss things. If you are not in the class Discord within the first week, email me.

4. Course progression (tentative)

The order below is the plan and is **tentative**. Pacing will be adjusted to the room, and the authoritative version of what has been covered is always the lecture notes posted in the repository. Announcements of any change go through Discord.

Unit	Subject	Contents
0	Foundations primer	Self-study. Notation, algebra, summation, functions, the calculus assumed from MATH 141.

Unit	Subject	Contents
1	Vectors	Vectors in $\mathbb{R}^2$ and $\mathbb{R}^n$; addition and scaling; the dot product, angle and projection; norms, metrics and Cauchy–Schwarz; span, basis, coordinates and dimension; linear independence; orthogonality and subspaces.
2	Computational foundations	Counting operations; asymptotic analysis and Big-O; floating-point arithmetic, machine precision and error; k-nearest neighbours; k-means clustering; the Gram–Schmidt process; matrix multiplication as an algorithm.
3	Matrices and linear systems	Linear systems from the coordinate problem; the matrix and the matrix–vector product; Gaussian elimination; matrix algebra, the transpose and the inverse; the determinant and the trace.
4	Linear maps and subspaces	Linear transformations and their matrices; kernel and image; rank–nullity; the four fundamental subspaces; change of basis.
5	Orthogonality and least squares	Orthogonal projection onto a subspace; the normal equations; least-squares fitting; the QR factorization.
6	Eigenvalues and applications	Eigenvalues and eigenvectors; diagonalization; an introduction to the singular value decomposition; applications including PageRank and image compression.

Complex numbers and quaternions appear as enrichment material alongside Unit 1 and are not examinable unless announced otherwise.

5. Grading

Component	Weight	Notes
Homework (5 assignments)	20%	Submitted via your GitHub repository.
Projects (3)	15%	Python implementations with a written report.
Midterm 1	20%	In class, in person.
Midterm 2	20%	In class, in person.
Final examination	25%	Cumulative; during the December 15–21 period.
Total	**100%**	

Extra credit will be offered during the semester and announced on Discord.

5.1. Grade scale

A+	97–100	B+	87–89.9	C+	77–79.9
A	93–96.9	B	83–86.9	C	73–76.9
A−	90–92.9	B−	80–82.9	C−	70–72.9
D+	67–69.9	D	63–66.9	D−	60–62.9
F	below 60				

5.2. Late work and make-ups

- **Homework and projects** are accepted up to 48 hours late with a 10% penalty per day. After 48 hours the grade is zero unless you have arranged something with me *in advance*.
- **Exams** are in class and cannot be made up except for a documented emergency, a religious observance (§9.7), or an approved accommodation (§9.3). Contact me as far in advance as you can; for an emergency, as soon as you are able.
- Talk to me early. A problem raised before a deadline has many more solutions than the same problem raised after it.

6. Attendance and Verification of Enrollment

Verification of Enrollment (VOE) — the first three weeks

CUNY requires me to take attendance at **every meeting during the first three weeks of the semester** and to submit Verification of Enrollment in CUNYfirst. For this section that means meetings 1 through 6: **September 1, 3, 8, 10, 15 and 17**.

If you do not attend during this window, you may be reported as not actively participating and **dropped from the course**. This is also how CUNY verifies enrollment for **financial aid**, so a missed verification can affect your aid disbursement. If something prevents you from attending during the first three weeks, email me immediately — before the fact if at all possible.

After the VOE window. Attendance is not itself a graded component of this course. What is graded is listed in §5. That said, this course moves quickly and builds on itself; the lecture notes are thorough but they are not a substitute for being in the room, and consistent absence reliably shows up in exam scores. If you must miss a class, get the notes, read them, and come to office hours with questions.

Lateness. Class starts at 12:10. Please arrive on time; if you do arrive late, come in quietly and take a seat rather than waiting outside.

7. Examinations

There are three examinations: two midterms and a cumulative final. All are taken **in class, in person**. There are no take-home exams. Programming is not assessed on exams (§1).

Exam	Date	Coverage
Midterm 1	Thursday, October 8 *(tentative)*	Units 1–2: vectors, inner products, norms, basis and span, and the computational material through Gram–Schmidt.
Midterm 2	Thursday, November 12 *(tentative)*	Units 3–4: linear systems, matrix algebra, determinants, linear maps and subspaces.
Final	During December 15–21	Cumulative, with emphasis on Units 5–6.

How exam dates are confirmed

The two midterm dates above are **tentative**. Each will be confirmed on Discord and in class **at least two weeks in advance**, and once confirmed it will not move. The final examination is scheduled by the College within the December 15–21 period; our meeting slots in that window are Tuesday, December 15 and Thursday, December 17. The official date and time will be announced as soon as the College publishes the final examination schedule.

Exams are designed so that a student who attends regularly, does the homework, and studies can do well. Questions will not be recycled verbatim from homework or review sheets: the goal is to test whether you understand the concepts, not whether you have memorized specific worked examples.

8. Homework, projects and submission

Five homework assignments and **three projects** across the semester. Both are submitted by committing and pushing to your own MATH 231 GitHub repository; a walkthrough covering git, GitHub accounts and directory structure is provided in week 1.

- **Homework** is a mix of hand computation and short proofs, with some Python. Show your reasoning — an unjustified correct answer earns partial credit at best. This is MQR 4, and it is graded.
- **Projects** ask you to implement, in Python, an algorithm already developed in lecture, and to submit a short written report: what you implemented, what you observed, and what the limits of the method are.
- **Collaboration** is encouraged on homework — talk to each other, form study groups, work problems together at the board. What you submit must be written by you, in your own words and your own code. Copying another student's solution, or letting yours be copied, is a violation of §9.1.

9. Course policies

9.1. Academic integrity

Academic dishonesty is prohibited in the City University of New York and is punishable by penalties, including failing grades, suspension, and expulsion. This course follows the CUNY Policy on

Academic Integrity as adopted by the CUNY Board of Trustees; the College's policy page is at `https://www.qc.cuny.edu/cetll/academic-integrity`.

Violations include plagiarism, copying another student's work, unauthorized collaboration on an exam, using prohibited materials during an exam, and submitting work you cannot explain as your own. Suspected violations are reported to the Office of Judicial Affairs. Queens College's Coordinator of Judicial Affairs is Emanuel Avila (emanuel.avila@qc.cuny.edu or studentconduct@qc.cuny.edu, 718-997-3971).

9.2. Generative artificial intelligence

You may use generative AI tools — ChatGPT, Claude, GitHub Copilot and the like — on **homework and projects**, to learn Python syntax, to debug, to get an explanation of a concept, or to check your understanding. I use these tools myself and I would rather teach you to use them well than pretend they do not exist. Two conditions:

1. **Disclose it.** Say so in a comment in your code, or in your project README: which tool, and what you used it for. A sentence is enough.
2. **Own it.** You must be able to explain every line you submit. I may ask you to.

Generative AI is not permitted on examinations. Submitting AI-generated work that you cannot explain, or failing to disclose AI use, is a violation of §9.1. Note also that these tools are confidently wrong about mathematics with some regularity — a habit of checking their output against what you know is part of MQR 5.

9.3. Students with disabilities and accommodations

The **Office for Students with Disabilities and Access (OSDA)** supports students with qualifying disabilities under the Americans with Disabilities Act by arranging reasonable accommodations. If you have previously received accommodations, believe you may have a disability, or have a temporary disability, contact OSDA:

Main office	Kiely Hall 104/108
Phone	718-997-5870
Email	OSDA@qc.cuny.edu
Testing office	Kiely Hall 220, 718-997-3775, osda.testing@qc.cuny.edu

Accommodations are not retroactive, so register sooner rather than later. Once OSDA sends me your accommodation letter, come talk to me — early in the semester if you can, and well before an exam — so we can make the arrangements without a rush. You are never required to disclose the nature of a disability to me.

Note: this office was formerly called the Office of Special Services (SPSV). Older documents may still use that name or the old email address.

9.4. Student wellness

College is hard, and it is harder when something else is going wrong. Anxiety, difficulty concentrating, loss, illness and stressful life events all affect academic performance, and none of them are things you have to handle alone. Confidential mental-health services are available on campus through **Counseling Services**, on the first floor of Frese Hall: 718-997-5420, CounselingServices@qc.cuny.edu, `https://www.qc.cuny.edu/cs/`.

A 24/7 crisis counselor is available: text "CUNY" to 741741.

If something is going on that is affecting your work in this class, you are welcome to tell me as much or as little as you want to. I would much rather hear about it early than reconstruct it from a transcript at the end of the semester.

9.5. Course withdrawals and incomplete grades

Deadlines for dropping and withdrawing are set by the College, not by me, and the grades `W`, `WN`, `WU` and `WA` have different consequences for your record and your financial aid. Check the deadlines yourself and talk to me before you withdraw: `https://www.qc.cuny.edu/provost/withdrawals-incomplete-grades/`.

An `INC` is possible only when most of the coursework is already complete and something outside your control prevents you from finishing; it is arranged in advance, never awarded automatically. CUNY's uniform grade glossary is at `https://www.cuny.edu/about/administration/offices/ur/resources/registrars-staff/CUNY_UNIFORM_GRADE_GLOSSARY091109.pdf`.

9.6. Repeating this course

Be aware of the College's policy on repeating courses and F-grade replacement: `https://www.qc.cuny.edu/aac/wp-content/uploads/sites/84/2022/04/Repeating-Courses-and-the-F-Grade-Replacement-Policy.pdf`. As the catalog description notes, students who fail or withdraw from MATH 231 multiple times may be prohibited from majoring in the sciences or mathematics.

9.7. Religious observance

Queens College accommodates religious observance, including for examinations. If a scheduled exam or deadline conflicts with a religious obligation, tell me as far in advance as possible and we will arrange an alternative. The College's policy is at `https://www.qc.cuny.edu/provost/religious-accommodations/`.

9.8. Grade appeals

If you believe a grade is in error, come to me first — most of these are arithmetic and take five minutes to fix. If we cannot resolve it, appeals go through the Undergraduate Scholastic Standards Committee: `https://www.qc.cuny.edu/saa/`.

9.9. Use of student work

Samples of student work are occasionally made available to accreditation professionals for periodic program reviews. Anonymity is assured under these circumstances.

9.10. Course evaluations

During the final four weeks of the semester you will be asked to complete an online course evaluation. All responses are completely anonymous; identifying information is erased once the evaluation has been submitted. I read these, and they change how I teach — please fill it out.

9.11. Non-discrimination and privacy

Queens College does not discriminate against students on the basis of pregnancy, childbirth or related conditions. Your educational records are protected under the Family Educational Rights and Privacy Act (FERPA); see `https://www.qc.cuny.edu/provost/guide-to-ferpa/`.

9.12. Emergency drills

A campus fire drill is scheduled for sometime during the week of September 28 – October 1, 2026. Two of our meetings fall in that window (Tuesday, September 29 and Thursday, October 1). If the alarm sounds during class, we evacuate Kiely Hall together by the nearest stairwell — not the elevator — and reassemble outside before returning. Any class time lost will be made up.

10. Getting help

In rough order of how quickly you will get an answer:

1. **The class Discord.** Ask in the open channel. Other students will often answer before I do, and answering someone else's question is one of the best ways to find out whether you actually understand something.
2. **Office hours**, Kiely Hall 331. Come with a specific question or come with nothing and we will find one. You do not need permission and you are not interrupting.
3. **Email**, per the policy on page 1.
4. **The Math Lab**, Kiely Hall 331 — free drop-in tutoring and a library of videos: `https://sites.google.com/view/qc-mathtutoring/home`.
5. **The QC Learning Commons**, Kiely 131 and 144: `https://www.qc.cuny.edu/academics/qclc/`.
6. **Form a study group.** Genuinely: the students who do well in courses like this one are disproportionately the ones who work together.

11. Important contacts

Office	Contact
Mathematics Department	718-997-5811 General: Math@qc.cuny.edu Registration: MathRegistration@qc.cuny.edu Placement: MathPlacement@qc.cuny.edu *Write to these addresses only — not to office staff individually.*
Office for Students with Disabilities and Access (OSDA)	Kiely Hall 104/108 718-997-5870 OSDA@qc.cuny.edu
OSDA Testing Office	Kiely Hall 220 718-997-3775 osda.testing@qc.cuny.edu
Academic Integrity / Judicial Affairs	Emanuel Avila, Coordinator of Judicial Affairs 718-997-3971 emanuel.avila@qc.cuny.edu or studentconduct@qc.cuny.edu
Counseling Services	Frese Hall, 1st floor 718-997-5420 CounselingServices@qc.cuny.edu `https://www.qc.cuny.edu/cs/` **24/7 crisis counselor: text "CUNY" to 741741**
Immigrant Student Support Services	718-997-3990 ImmigrantSupport@qc.cuny.edu `https://www.qc.cuny.edu/immi/`
Technology Services (ITS)	I-Building 200 718-997-4444 support@qc.cuny.edu Support tickets: `https://support.qc.cuny.edu/support/home`
Security Office	Main Gate 718-997-5911 / 718-997-5912
Audio Visual (media equipment)	718-997-5960
Human Resources	Kiely Hall 163 718-997-4455 Ohr.Employeeservices@qc.cuny.edu
Payroll Office	Kiely Hall 151 718-997-5765 Ohr.Payroll@qc.cuny.edu
Provost — course processes and policies	`https://www.qc.cuny.edu/provost/course-processes/`
CUNY academic calendar	`https://www.cuny.edu/academics/academic-calendars`

12. Meeting schedule

Twenty-eight regular lecture meetings, September 1 through December 10. Two scheduled meetings are lost to the College calendar. **This schedule is subject to change**; changes are announced on Discord.

#	Date	Day	#	Date	Day
1	Sep 1	Tue	15	Oct 22	Thu
2	Sep 3	Thu	16	Oct 27	Tue
3	Sep 8	Tue	17	Oct 29	Thu
4	Sep 10	Thu	18	Nov 3	Tue
5	Sep 15	Tue	19	Nov 5	Thu
6	Sep 17	Thu	20	Nov 10	Tue
7	Sep 22	Tue	21	Nov 12	Thu
8	Sep 24	Thu	22	Nov 17	Tue
9	Sep 29	Tue	23	Nov 19	Thu
10	Oct 1	Thu	24	Nov 24	Tue
11	Oct 6	Tue	–	*Nov 26*	*Thu*
12	Oct 8	Thu	25	Dec 1	Tue
–	*Oct 13*	*Tue*	26	Dec 3	Thu
13	Oct 15	Thu	27	Dec 8	Tue
14	Oct 20	Tue	28	Dec 10	Thu

Date	What happens
Sep 1–17 (meetings 1–6)	VOE attendance window (§6)
Sep 28 – Oct 1	Campus fire drill, exact day to be announced (§9.12)
Thursday, October 8	Midterm 1 *(tentative — confirmed two weeks ahead)*
Tue, Oct 13	**No class** — College follows a Monday schedule
Thursday, November 12	Midterm 2 *(tentative — confirmed two weeks ahead)*
Thu, Nov 26	**No class** — College closed, Thanksgiving
Thu, Dec 10	Last regular lecture
Dec 15–21	Final examination period